Behaviorally Validated Corpus Mutation for Retrieval Robustness Evaluation

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***Abstract*—Synthetic retrieval failures are easy to generate with large language models, but it remains unclear when a generated corpus mutation counts as a genuine retrieval failure rather than merely a plausible-looking one. We argue that semantic validity, behavioral effectiveness, and cross-retriever transferability are three distinct properties of a synthetic retrieval failure, and that a trustworthy benchmark-construction pipeline must measure each separately rather than assuming one implies the next. We instantiate this argument with a four-stage pipeline — Generate, Semantic Validate, Behaviorally Validate, Transfer — applied to five corpus-mutation operators (missing, hard negative, conflict, boundary, noise) on the RAGFailBench corpus, and evaluate three distinct properties of synthetic retrieval failures. First, automatic semantic validation and independent human audit jointly establish that the corrected mutation operators satisfy their intended semantic definitions, with 99.6 percent overall human-audit acceptance. Second, semantic validity does not imply behavioral effectiveness: our noise operator is semantically valid in effectively 100 percent of cases but behaviorally effective in under 1 percent, and a topically valid non-LLM hard-negative baseline (96 percent human-audit ok) is far less behaviorally effective than LLM-generated hard negatives. Third, behavioral effectiveness does not imply transferability: hard negative and conflict mutations accepted on three construction retrievers still challenge two held-out retrievers in only about one quarter of cases. These results argue for measuring corpus-mutation quality along three separate axes rather than collapsing it into a single score.**

***Index Terms*—retrieval evaluation, corpus mutation, hard negatives, behavioral validation, retrieval-augmented generation, benchmark construction**

I. Introduction

Synthetic retrieval failures are increasingly easy to generate — an LLM can rewrite a passage into a plausible hard negative or a contradicting fact in seconds — but it remains unclear when a generated mutation should actually count as a retrieval failure. Semantic plausibility alone does not establish retrieval difficulty: a mutation an annotator or LLM judge rates as well-formed can still leave every retrieval metric unchanged. And even measurable degradation on the retrievers used to build the benchmark does not establish that the effect generalizes: a mutation that reliably fools one retriever family can be invisible to another. This paper's thesis is that semantic validity, behavioral effectiveness, and cross-retriever transferability are three distinct properties of a synthetic retrieval failure, and a trustworthy benchmark-construction pipeline must measure each of them separately rather than assuming one implies the next.

Existing resources address this problem only partially. Static benchmarks such as BEIR and MS MARCO fix a corpus and relevance judgments at release time, so they cannot be regenerated as an organization's knowledge base evolves — the setting that matters for enterprise RAG. Synthetic and LLM-generated hard negatives are typically validated for semantic well-formedness only: a judge confirms a mutated passage looks like a plausible distractor, and that judgment is then treated as sufficient grounds to include the item in a benchmark. Neither line of work asks whether an accepted mutation actually moves a retrieval metric, or whether the effect it produces is specific to the retriever that was used to construct it.

We build a four-stage pipeline, Generate → Semantic Validate → Behaviorally Validate → Transfer (Fig. 1), that treats each of these three properties as something to be measured, not declared. Five corpus-mutation operators — missing, hard negative, conflict, boundary, and noise — are applied one at a time to clean seeds, so that a mutation's effect can be attributed to a single, well-defined failure rather than a compound of several. Each candidate mutation passes through an automatic semantic validator (Definition 1), then through a behavioral filter that checks, on three construction retrievers (BM25, a dense bi-encoder, and a hybrid), whether the mutation actually produces a failure-specific retrieval effect (Definition 2). Only mutations accepted at this stage are then evaluated for transfer on two held-out dense retrievers, E5-large and GTE-large, that never participate in construction or filtering (Definition 3).

This design lets us ask three separate research questions, and running the pipeline on the RAGFailBench corpus yields one headline finding per question. RQ1: can controlled corpus mutations be shown, rather than assumed, to satisfy their intended failure definitions? F1 — automatic semantic validation and independent human audit provide complementary evidence of mutation validity; the current operators achieve 99.6% overall human-audit acceptance. A development-time boundary defect — caught by human audit at 0/50 ok despite 100% automatic SVR, then fixed — further illustrates why independent audit is useful (§V-A). RQ2: which semantically valid mutations actually induce meaningful retrieval degradation? F2 — semantic validity does not imply behavioral effectiveness: the noise operator is semantically valid in effectively 100% of cases but behaviorally effective in under 1%. RQ3: to what extent do behaviorally valid mutations — accepted on con-

struction retrievers — transfer to held-out retrievers that never saw them? F3 — behavioral effectiveness does not imply universal transferability: hard negative and conflict mutations, both accepted on construction retrievers, still only challenge held-out retrievers in ≈24–25% of cases, with non-transfer concentrated in instances where the held-out retriever ranks the mutated-in chunk inside its own top-20 without disturbing the gold passage's rank, rather than in instances where the mutation never reaches the held-out candidate set (§V-C) — evidence that retrieval failures contain both retriever-specific and cross-retriever components, not evidence that generation failed. A fourth, supporting result (§V-E) extends F2: a non-LLM, BM25-mined hard-negative baseline is rated almost as topically valid as our LLM-generated hard negatives by human audit, yet is far less behaviorally effective, in a setting where the confound of noise's unusual construction cannot explain the gap.

Our contributions are: (1) we formalize semantic validity, behavioral effectiveness, and cross-retriever transferability as distinct properties of a synthetic retrieval failure (Definitions 1–3), rather than treating one as a proxy for the others; (2) we introduce a behavioral validation framework that accepts a mutation only when it produces a predefined, failure-specific retrieval effect across multiple construction retrievers, formally distinguished from semantic validation; (3) we separately evaluate transfer on two held-out retrievers that never participate in construction, showing that construction-time behavioral validity does not guarantee cross-retriever generalization; and (4) we instantiate the framework with five controlled, retrieval-level corpus-mutation operators and report two independent semantic–behavioral dissociations: noise, and a BM25-mined hard-negative baseline that is nearly as topically valid as our LLM-generated hard negatives by human audit yet substantially less behaviorally effective. To be precise about scope, the contribution is not the use of an LLM to generate hard negatives, nor the use of retrieval metrics to check generated data — both appear in prior work (§II-C) — but the explicit separation of semantic validity, construction-time behavioral effectiveness, and held-out transferability within a single, reproducible corpus-mutation evaluation framework.

II. Related Work

A. Retrieval Benchmarks

Large-scale supervised and zero-shot retrieval benchmarks such as MS MARCO [12] and BEIR [13] established heterogeneous test beds for retriever comparison, alongside open-domain QA sets such as Natural Questions [14] and HotpotQA [15], the knowledge-intensive task suite KILT [16], and the multilingual benchmark MIRACL [17]. Each release fixes its corpus and relevance judgments at publication time; none is designed to be regenerated as a target knowledge base evolves, which is the deployment setting this paper targets.

B. Sparse and Dense Retrieval

Classical sparse retrieval, particularly BM25 [1], [2], remains a strong lexical baseline and serves as one of our construction retrievers. Dense bi-encoders — including DPR [4], ANCE [7], and Contriever [8] — established neural retrieval based on learned representations, while E5 [9], GTE [10], and BGE [11] provide the embedding-model families used in our construction and held-out evaluation. We additionally combine BM25 and BGE via reciprocal rank fusion (RRF) [3] to obtain a hybrid construction retriever. Late-interaction models such as ColBERT [5] and ColBERTv2 [6] (which also introduced the LoTTE benchmark) represent another retrieval family, not evaluated here — a gap we flag directly in our transfer-generalization limitation (§VII).

C. Hard Negative Construction

Hard-negative mining for contrastive training and evaluation ranges from in-batch and cross-batch negatives (RocketQA [18]) to approximate-nearest-neighbor mining (ANCE [7]) and unsupervised domain-adaptive pseudo-labeling (GPL [21]). Document expansion via query prediction [19] and adversarial-passage poisoning of retrieval corpora [20] are related but distinct constructions: the former augments documents rather than injecting failures, and the latter targets attack rather than evaluation. LLM-based synthetic query and hard-negative generation — InPars [22], InPars-v2 [24], and Promptagator [23] — is the most directly related line of work to this paper's Hard Negative operator (§III-A); like this paper, these methods use an LLM to generate query-document or negative-document pairs, but validate outputs for downstream retrieval quality (e.g., fine-tuning gains) rather than for a separately measured, failure-specific behavioral acceptance criterion prior to benchmark inclusion.

D. LLM-based Data Generation

Beyond retrieval, using LLMs to generate training or evaluation data has been studied broadly: Self-Instruct [25] and Unnatural Instructions [26] generate instruction-tuning data from a small seed set; ZeroGen [27] and symbolic knowledge distillation [28] generate task-specific or commonsense training data directly from a teacher LLM without human-labeled examples. These methods share this paper's reliance on an LLM as a generator, but, like the hard-negative generation work in §II-C, validate outputs against downstream task performance rather than against a standalone behavioral-effect criterion measured before the generated item is accepted into a benchmark.

E. RAG and Retrieval Robustness Evaluation

RAG evaluation frameworks such as RAGAS [29] and ARES [30] score end-to-end generation quality (faithfulness, answer relevance) rather than isolating the retrieval stage. Broader RAG benchmarks — RGB [31], CRUD-RAG [32], and RAGBench [33] — probe robustness to noisy or counterfactual context at the level of the full pipeline. Retrieval-specific robustness work is closer in spirit to this paper: certifiable robustness to retrieval corruption [34], entity-based knowledge conflicts in QA [35] (directly relevant to this paper's Conflict operator, §III-A), simple entity-centric queries

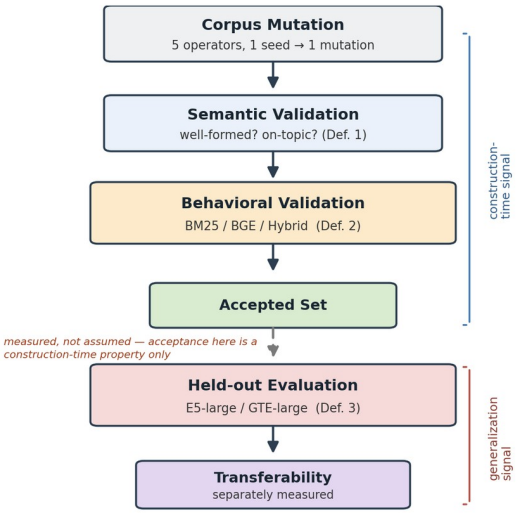


Fig. 1. Behavioral validation (top bracket) measures a construction-time signal on BM25/BGE/Hybrid; held-out evaluation (bottom bracket) separately measures a generalization signal on E5/GTE. Acceptance into the benchmark does not by itself imply transfer — the dashed arrow marks where the two signals are deliberately decoupled.

that challenge dense retrievers [36], and training with harder negatives to improve robustness [37]. This paper differs from all of the above in scope: it stays upstream of generation (§VII) and treats validity, behavioral effectiveness, and cross-retriever transfer as three separately measured properties of a corpus mutation, rather than reporting a single end-to-end robustness score.

III. Method

The pipeline has four stages — Corpus Mutation, Semantic Validation, Behavioral Validation, Transfer Evaluation — applied to clean seeds merged from prior RAGFailBench stability runs (531 unique seeds in the main run reported here). Each seed yields exactly one mutated corpus state per failure type; failures are never compounded within a single instance.

We formalize the three properties measured by the pipeline, since each is reported as a separate quantity throughout §V rather than collapsed into one benchmark-quality score.

Definition 1 (Semantic Validity). A mutation is semantically valid if it passes the operator-specific automatic semantic checks for topical well-formedness, non-degeneracy, and failure intent (structural rule checks for every operator; an additional LLM-judge pass for hard negative and conflict, where generation quality is subjective). The validity procedure itself — not every individual mutation — is independently assessed via a stratified human audit (§III-B).

Definition 2 (Behavioral Validity). A semantically valid mutation is behaviorally valid if it produces a predefined, failure-specific retrieval effect — defined per operator in §III-C — on a majority of the construction retrievers. Behavioral validity is necessary for benchmark inclusion; semantic validity alone is not sufficient.

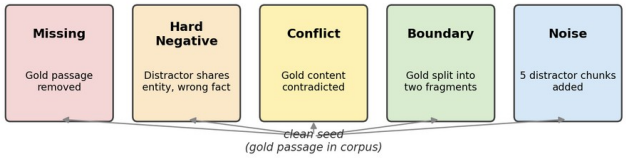


Fig. 2. The five corpus-mutation operators, applied one at a time to a clean seed.

Definition 3 (Transferability). A behaviorally valid mutation is transferable to the extent that its retrieval effect (challenge rate, ΔMRR) reproduces on retrievers that did not participate in construction or filtering (§III-D). Transferability is measured, not assumed, for every accepted operator.

A. Corpus Mutation

- **Missing** — rule-based removal of the gold passage. Deterministic; used as a negative control (§III-C).
- **Hard Negative** — an LLM rewrites a distractor sharing the gold passage's entity/relation but asserting a different, plausible fact (prompt-only). A Random Hard-Negative baseline mines a topically related chunk via BM25, no LLM, for head-to-head comparison.
- **Conflict** — an LLM rewrites the gold passage's content to contradict it while remaining topically on-target.
- **Boundary** — rule-based split of the gold passage into two pieces (A/B).
- **Noise** — five additional distractor chunks are added (corpus-mined / LLM-generated, BM25-filtered) without altering the gold passage.

B. Semantic Validation

Deterministic structural and answer-leakage checks — gold removed, split pieces non-empty, answer not restated — apply to every operator. For hard negative and conflict, where generation quality is subjective and rules alone cannot verify it, we additionally require an LLM judge (gpt-oss-120b, temperature 0) to return three booleans — topical relevance, naturalness, and failure intent — before a mutation passes; missing, boundary, and noise are validated by rules alone. If the judge's response is truncated or unparsable after one retry, validation fails open on the already-passed rule checks rather than rejecting the mutation outright, since the rule checks alone already establish structural validity. This was the active validation policy throughout the experimental runs reported in this paper. Auditing every LLM-judge call issued during the main run, this fail-open path was never exercised: of 1,019 primary-run judge calls (hard negative and conflict), 982 returned pass and 37 returned reject, with zero fail-open outcomes; including the Random Hard-Negative baseline (§V-E), the full run totals 1,550 judge calls (1,301 pass, 249 reject, zero fail-open).

For the independent human audit, we sample 50 instances per failure type with a fixed seed (seed=0), yielding 250

audited instances in total. A single annotator assigns one of five labels — ok, topical_fail, answer_leak, unnatural, or other — against a fixed codebook. The audit reports 249/250 (99.6%) ok, with the sole miss a Random Hard-Negative instance (§V-E) labeled topical_fail. Because the audit uses a single annotator, we do not report inter-annotator agreement (§VII).

C. Behavioral Validation

A semantically valid mutation is accepted only if it produces a failure-specific retrieval effect on a majority (≥ 2 of 3) of the construction retrievers (BM25, BGE-base, BM25+BGE hybrid via RRF, k=60). Criteria are failure-specific: Hard Negative — outranks gold, top-5 crossing, or $\Delta\text{MRR} \geq 0.10$; Conflict — the conflicting chunk enters the top-5 and either outranks gold or $\Delta\text{MRR} \geq 0.10$; Noise — top-5 entry and gold-rank drop ≥ 3 or top-5 crossing; Boundary — split relevance. Missing is excluded (deterministic; operation success rate 1.0), serving as a negative control rather than a validated item. Behavioral validation uses construction retrievers only — the held-out retrievers never participate. §V-B reports an offline sensitivity check on the $\Delta\text{MRR} \geq 0.10$ and $\geq$2/3-agreement choices made here.

D. Transfer Evaluation

Accepted mutations are evaluated on two held-out dense retrievers, E5-large and GTE-large. For a mutation m we define $\Delta\text{MRR}(m) = \text{MRR}_{\text{clean}}(m) - \text{MRR}_{\text{mutated}}(m)$ and $\Delta\text{Recall@5}(m) = \text{Recall@5}_{\text{clean}}(m) - \text{Recall@5}_{\text{mutated}}(m)$, so a positive Δ denotes degradation after mutation on either metric. An instance is challenged, $CR(m) = \mathbb{1}[\Delta\text{MRR}(m) > 0 \vee \Delta\text{Recall@5}(m) > 0]$, if either metric degrades; challenge rate is the share of accepted instances with $CR(m) = 1$, and we always report it alongside both Δ values per failure so which metric is driving a given challenge rate is visible (§V-C). §V-C reports both an offline check on whether this aggregate rate is stable across the semantic and behavioral filtering stages and a bucket-level breakdown of why non-transferring instances do not transfer, and §V-E repeats the pipeline for the Random Hard-Negative baseline.

IV. Experimental Setup

Dataset: 531 unique clean seeds merged from four RAG-FailBench stability runs; up to $5 \times 531 = 2{,}655$ candidate mutations before filtering (2,322 realized). Generation: hard-negative and conflict distractors are produced by an LLM (gpt-oss-120b, temperature 0); missing and boundary are purely rule-based. Semantic validation combines structural rule checks — applied to every operator (e.g., gold removed, split pieces non-empty, answer not restated) — with an LLM-judge pass (natural, on-topic, correct failure intent) applied only to hard negative and conflict, where generation quality is subjective and rules alone cannot verify it; noise and boundary rely on rules only. Retrievers: BM25, BGE-base-en-v1.5, and BM25+BGE hybrid (RRF, k=60) for construction; E5-large-v2 and GTE-large held out; retrieval cutoff k=20 throughout, so held-out rank changes beyond depth 20 are not observed (§VII). Metrics: Recall@5, Recall@20, MRR, gold-rank displacement, top-5 rank-crossing rate, paired against the clean run. Statistics: bootstrap confidence intervals and two-sided significance tests use 1,000 resamples with a fixed seed; large-effect comparisons are reported without CIs where unambiguous. Protocol: all behavioral acceptance criteria and thresholds (§III-C) are fixed in code prior to and independent of any held-out evaluation, and held-out retrievers are applied only after behavioral acceptance is finalized — no threshold in this paper was tuned against E5/GTE results. The threshold-sensitivity (§V-B) and failure-taxonomy (§V-C) analyses are offline recomputations from these same frozen rankings and do not re-run generation, embedding, or retrieval; human-audit sampling for both follows the same fixed seed=0 protocol as §III-B.

Code and Reproducibility. The implementation, mutation operators, validation procedures, experimental configurations, and evaluation scripts are publicly available at https://github.com/taixingbi/corpus-mutation-retrieval-robustness. The repository also includes experimental reports and audit scripts supporting the reported results.

V. Results

We organize results around the three research questions posed in §1: semantic validity (RQ1, §V-A), behavioral effectiveness (RQ2, §V-B), and cross-retriever transferability (RQ3, §V-C). Sections V-B–V-E also report supporting analyses of threshold sensitivity, aggregate transfer-rate interpretation, non-transfer taxonomy, and hard-negative generation baselines, that qualify, respectively, RQ2, RQ3, and the value of LLM-based generation. All results below reflect the current, corrected version of every operator (§V-A reports the boundary answer-leak defect we found and fixed during development as a case study, not a live issue). Hard negative and conflict are the paper's primary challenge operators (BER 70–76%, hundreds of accepted instances each, consistent ΔMRR-driven degradation at construction time and on both held-out retrievers). Boundary is retained as a diagnostic fragmentation operator, marked † throughout: its behavioral signal is real (BER 65.7%) but is carried almost entirely by ΔRecall@5 rather than ΔMRR (§V-C, §VI), which is itself informative about what different retrieval metrics capture. Noise serves as a controlled semantic–behavioral counterexample: semantically valid in effectively every case but behaviorally effective in under 1%, and currently yields too few behaviorally accepted instances ($n = 5$) to serve as a practical benchmark operator without further tuning (§VII). Missing remains a deterministic negative control outside the BER framework.

A. RQ1 — Semantic Validity

RQ1: Can controlled corpus mutations be shown, rather than assumed, to satisfy their intended failure definitions?

Finding 1: automatic semantic validation and independent human audit provide complementary evidence of mutation validity. Table I reports automatic Semantic Validity Rate (SVR)

TABLE I
AUTOMATIC SVR VS. HUMAN AUDIT

Failure	SVR	Human	Notes
Missing	100%	100%	Control
Hard Neg.	94.3%	98%	LLM 23/23; Rand. 26/27
Conflict	98.3%	100%	1 reclassified on review
Boundary †	100%	100%	fixed post answer-leak
Noise	100%	100%	—

n=50/failure. Overall human ok rate: 99.6%. †Boundary: an earlier version failed human audit (0/50, answer_leak) despite 100% automatic SVR; the split logic was fixed and the operator now passes both checks (§VI).

TABLE II
BER AND PAIRED ΔMRR

Failure	Valid	Effective	BER	ΔMRR
Hard Neg.	461	350	75.9%	0.132
Conflict	521	365	70.1%	0.128
Boundary †	251	165	65.7%	−0.024 n.s.
Noise	531	5	0.9%	0.058

Majority vote, 3 construction retrievers; paired ΔMRR on bge_base. †Boundary's ΔMRR is not significantly different from zero (n.s.) on any construction retriever; its BER reflects the split-relevance criterion, a different measurement than ΔMRR (§VI).

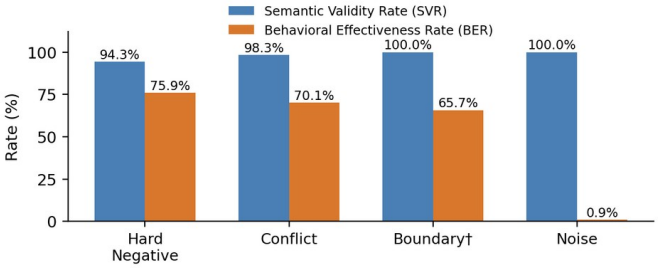


Fig. 3. Semantic Validity Rate vs. Behavioral Effectiveness Rate, by failure type. Noise is the paper's central dissociation: SVR ≈ 100% but BER < 1%, i.e. semantic validity does not imply behavioral effectiveness.

alongside a fresh 50-item human audit per failure type, on the current, corrected version of every operator: both are near-ceiling today (overall human ok rate: 99.6%). The value of the independent audit component is visible in the development history that produced this agreement: an initial version of boundary passed automatic review at 100% (251/251) while a human audit on that version rated it 0/50 ok, every sampled item flagged answer_leak — the split logic sometimes let one fragment retain the full gold answer, a defect automatic review had no way to detect since a leaked answer is by construction semantically coherent and topically on-target. We fixed the split logic accordingly; the boundary † row below reflects the corrected operator, now audited clean. Conflict shows a smaller instance of the same principle: one of 50 initial samples was first flagged answer_leak, then reclassified ok on review because the gold value appeared only as incidental backstory rather than a restated answer — a reminder that human audit outcomes themselves benefit from a documented review pass, not a single label.

B. RQ2 — Behavioral Effectiveness

RQ2: Which semantically valid mutations actually induce meaningful retrieval degradation?

Finding 2: semantic validity does not imply behavioral effectiveness. Table II reports Behavioral Effectiveness Rate (BER) and paired ΔMRR on bge_base. Noise is the clearest evidence for this finding: SVR = 100% but BER = 0.94% (5/531). We treat this as the paper's most informative result precisely because the operator is otherwise unremarkable — well-formed, on-topic distractors that a semantic reviewer would wave through almost never move retrieval outcomes, which is exactly the gap behavioral validation exists to catch. Hard negative (75.9%) and conflict (70.1%) show the highest BER; boundary† (65.7%) is also substantial, but by a different criterion — its ΔMRR is near zero and not statistically significant on any construction retriever ($p = 0.36$ bm25, 0.12 bge_base, 0.45 hybrid), so its behavioral acceptance here reflects split relevance (Definition 2), not a change in ranked position. All hard_negative/conflict ΔMRR effects, by contrast, are significant ($p \approx 0$, bootstrap) across all three construction retrievers.

Sensitivity to the behavioral acceptance threshold. Because Definition 2's ΔMRR ≥ 0.10 cutoff is a free parameter, we recompute construction-time acceptance from frozen construction-retriever rankings and evaluate the resulting accepted sets using frozen held-out rankings, under three ΔMRR thresholds (0.05, 0.10, 0.15) at the ≥2/3 construction-retriever agreement used throughout the paper, plus a stricter 3/3-agreement configuration at ΔMRR ≥ 0.10 (Table III). Varying the ΔMRR threshold alone leaves acceptance and downstream rates unchanged: all three configurations accept the same 885 mutations, with identical pooled BER (50.2%) and mean held-out challenge rate (32.3%). This is because accepted hard-negative and conflict instances already satisfy the alternative outranking or top-5-crossing criteria (§III-C), so the ΔMRR cutoff is non-binding across the tested range. Requiring unanimous (3/3) construction-retriever agreement instead of majority (≥2/3) is the criterion that matters: it shrinks the accepted set to 737 and lowers pooled BER to 41.8%, while the held-out challenge rate changes only modestly (32.3%→30.2%). We read this as evidence that the reported results are not an artifact of the specific 0.10 threshold, though they do depend on the majority- vs. unanimous-agreement choice; the Original row matches the frozen Table II/IV numbers used throughout the paper.

C. RQ3 — Cross-retriever Transferability

RQ3: To what extent do behaviorally valid mutations transfer to held-out retrievers that never participated in construction?

Finding 3: behavioral effectiveness does not imply universal transferability. Table IV and Fig. 4 report held-out CR, ΔMRR, and ΔRecall@5 for the accepted set, using the definitions in §III-D. Hard negative and conflict — already accepted on construction retrievers — still only challenge E5/GTE in ≈24–25% of cases, with ΔMRR ≈ 0.08–0.10 and only a small accompanying ΔRecall@5 (≤0.01); for these two operators, CR is effectively an MRR-degradation signal. We

TABLE III
BEHAVIORAL THRESHOLD SENSITIVITY (OFFLINE)

Config.	ΔMRR	Agree.	Acc.	BER	CR
Lenient	≥0.05	≥2/3	885	50.2%	32.3%
Original	≥0.10	≥2/3	885	50.2%	32.3%
Strict	≥0.15	≥2/3	885	50.2%	32.3%
Consensus	≥0.10	3/3	737	41.8%	30.2%

Offline acceptance recomputation from frozen construction-retriever rankings, followed by evaluation on frozen held-out rankings. ΔMRR threshold applies to hard negative and conflict only (noise/boundary criteria are fixed; missing is excluded from BER, §III-C). CR is pooled mean held-out challenge rate over E5-large and GTE-large. Original matches the frozen main-run numbers reported in Tables II and IV.

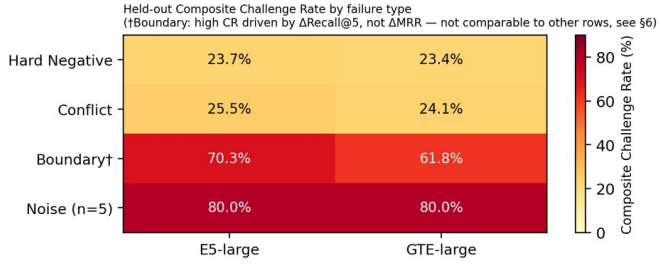


Fig. 4. Held-out Composite Challenge Rate by failure type and retriever. CR aggregates Recall@5 degradation and/or ΔMRR (§III-D) into one number and should not be read as a common severity scale across mutation types — boundary's high CR reflects its Recall@5 component, not ΔMRR; see Table IV for the separate breakdown and §VI for discussion.

TABLE IV
HELD-OUT COMPOSITE CHALLENGE RATE

Failure	n	E5 CR	GTE CR	ΔMRR E/G	ΔR@5 E/G
Hard Neg.	350	23.7%	23.4%	0.092/0.082	0.006/0.006
Conflict	365	25.5%	24.1%	0.097/0.075	0.005/0.011
Boundary †	165	70.3%	61.8%	0.017 n.s. / −0.066*	0.339/0.291
Noise	5	80.0%	80.0%	0.073/0.164	0.000/0.000

$CR = 1$ if ΔMRR>0 or ΔRecall@5>0 (§III-D; not a common severity scale across operators), reported as E5/GTE (positive Δ = degradation after mutation). †Boundary's ΔMRR is not significant on E5 (n.s.; CI crosses zero); *the GTE value is significantly negative, meaning mutated MRR is marginally higher than clean there. Boundary's ΔRecall@5 is an order of magnitude larger than any other operator on both retrievers — its CR is driven almost entirely by this Recall@5 degradation, since its ΔMRR rarely satisfies the CR condition (§VI).

TABLE V
HELD-OUT NON-TRANSFER TAXONOMY (ACCEPTED SET)

Failure	Retr.	Chall.	Never Top-20	Top-20 gold stable	n
Hard Neg.	E5	23.7%	0.6%	75.7%	350
Hard Neg.	GTE	23.4%	0.3%	76.3%	350
Conflict	E5	25.5%	0.0%	74.5%	365
Conflict	GTE	24.1%	0.0%	75.9%	365

Offline recomputation from saved held-out top-20 rankings; ranks beyond depth 20 are not observed (§VII). Percentages are of each row's own accepted, per-retriever population (n) and sum to 100% within a row (Challenged + Never Top-20 + Top-20-gold-stable) — not a share of a pooled non-transfer count.

read this not as a failure of generation, but as evidence that retrieval failures contain both a retriever-specific component (captured at construction time, §V-B) and a smaller, cross-retriever component (captured here). Boundary†'s CR (62–70%) is driven almost entirely by ΔRecall@5, not ΔMRR: Recall@5 drops by 0.29–0.34 on both held-out retrievers — the largest degradation of any operator on this metric — while ΔMRR stays near zero and not significant on E5, and is significantly negative on GTE (-0.066), meaning mutated-corpus MRR is marginally higher than clean there. We discuss what this pattern implies about metric choice in §VI. Noise has only 5 accepted instances (§V-B); its 80% figure is descriptive only and insufficient for a transfer conclusion at this sample size.

Boundary challenge decomposition. Among 165 behaviorally accepted boundary mutations, E5 records 95 Recall@5-only, 6 MRR-only, and 15 joint challenges (116 challenged total, matching the 70.3% CR in Table IV); GTE records 86, 4, and 12, respectively (102 challenged total, matching 61.8%). The boundary operator replaces one relevant gold passage with two relevant fragments. Retrieving only one fragment in the top-5 reduces fragment-level Recall@5 from 1.0 to 0.5, even when the first relevant fragment maintains or improves its rank. Thus, boundary primarily exposes evidence-fragment coverage degradation rather than first-relevant ranking degradation. Fragment coverage does not, by itself, establish that both fragments are necessary for a correct generated answer.

Why do behaviorally accepted mutations fail to transfer? Using the saved held-out top-20 rankings, we categorize each accepted hard-negative and conflict mutation's held-out outcome into three buckets: challenged ($CR(m) = 1$, as defined in §III-D); entered the held-out top-20 but the gold passage's own rank was unchanged (non-transfer, mutation visible); and never entered the held-out top-20 (non-transfer, mutation not visible). Table V reports this breakdown for the two primary challenge operators. For both, non-transfer is dominated by the gold-stable bucket — 74.5–76.3% of each operator's own accepted, per-retriever population — rather than by the mutation failing to reach the held-out candidate set at all (≤0.6%). In other words, when a behaviorally accepted hard-negative or conflict mutation does not transfer, it is almost always because the held-out retriever still ranks the gold passage correctly despite retrieving the mutated chunk into its own top-20, not because the mutation is invisible to the held-out retriever. This supports the retriever-specific-ranking-sensitivity reading of F3 above and in §VI, rather than a failure of mutation construction. (Boundary shows a different pattern, consistent with its ΔRecall@5-driven CR in Table IV: most of its accepted instances are challenged outright rather than falling into either non-transfer bucket, so we do not include it in Table V.)

D. Diagnostic Note: Aggregate Transfer Rate Is Not Meaningful in Isolation

A natural summary statistic — held-out challenge rate pooled across all failures, at each filtering stage — moves only slightly under semantic filtering (Raw 35.3% → +Semantic 35.2%, n=2,332→2,295) and then drops to 32.3% (n=885) once behavioral filtering is applied. We do not read this drop as evidence that behavioral validation reduces — or improves — transferability: behavioral filtering removes 526/531 noise

instances and re-weights the accepted pool toward hard negative and conflict (≈24–25% CR each), away from boundary's much higher but mechanistically distinct CR (62–70%, §V-C). This is a composition effect, not a treatment effect, and it is exactly why we report CR per failure (Table IV) rather than pooled: behavioral validation selects for construction-time retrieval effects; transferability remains a separate empirical property that must be measured on held-out retrievers, as we do in §V-C and above.

E. Supporting Analysis: LLM-Generated vs. Random Hard Negatives

This section provides a second, independent test of the paper's RQ2 thesis. §V-B showed the dissociation using noise, an operator that adds content without touching the gold passage; here we test whether the same dissociation appears within a single operator (hard negative) when only the generation source changes. We compare the LLM-generated Hard Negative operator against a Random Hard-Negative baseline (BM25-mined, no LLM) under identical semantic and behavioral filters. Unlike Table IV, which evaluates transfer only over the behaviorally accepted set, Table VI reports challenge rates over the semantic-valid population, so that generation sources can be compared before behavioral selection has a chance to differentially filter them; this is why LLM HN's E5/GTE challenge rate here (32.5%/33.4%) is higher than its Table IV figure (23.7%/23.4%), which is computed on the smaller, already-accepted subset.

Supporting evidence for F2: a mutation can be judged almost as topically valid as an LLM-generated one by human audit, and still be far less behaviorally effective. Human audit (§V-A) rates Random HN 26/27 (96%) topically ok — nearly on par with LLM HN's 23/23 (100%). Table VI shows that this near-parity in semantic validity does not carry over to behavioral effectiveness: on the semantically valid population, challenge rate is 4–34% (LLM) vs. 4–10% (Random) depending on retriever, ΔMRR is roughly 20–30× larger for LLM (0.06–0.13 vs. 0.004–0.007), and the gap persists on held-out retrievers (E5: 32.5% vs. 2.5%; GTE: 33.4% vs. 5.3%). Naive corpus-similarity mining produces a topically plausible hard negative that is nevertheless nearly behaviorally inert — the same pattern as noise (F2, §V-B), now demonstrated within a single operator by varying only the generation source, which rules out an explanation specific to noise's unusual construction. Read together with §V-C (F3): LLM generation is substantially more effective than our BM25-mined baseline at producing behaviorally challenging hard negatives — a claim specific to this baseline and this acceptance pipeline, not a general claim that LLM-generated negatives dominate retrieval-mined negatives as a class — but generation alone remains insufficient to guarantee transfer — all three properties must still be measured separately, which is this paper's central argument.

TABLE VI
LLM-GENERATED VS. RANDOM HARD NEGATIVES

Metric	LLM HN	Random HN
n (semantic-valid)	461	319
CR — bge_base	34.3%	4.1%
ΔMRR — bge_base	0.132	0.0042
CR — E5 (held-out)	32.5%	2.5%
CR — GTE (held-out)	33.4%	5.3%
Human audit ok	23/23 (100%)	26/27 (96%)

LLM-generated vs. Random hard negatives after semantic validation, evaluated under identical behavioral and held-out evaluation protocols. Human-audit rows repeat §V-A for direct comparison.

VI. DISCUSSION

Why is noise semantically easy but behaviorally hard? The behavioral criterion requires a distractor to enter the top-5 and displace gold by $\geq$3 ranks or cause a rank crossing — a stronger bar than "plausible distractor," which is what the semantic validator checks. Five added, topically related chunks rarely dominate a well-indexed gold passage, so noise passes semantic review almost automatically while failing the behavioral bar almost as automatically — the clearest demonstration that semantic validity and behavioral effectiveness are separate properties requiring separate measurement (RQ2).

Why do behaviorally effective mutations transfer only partially? Construction-time acceptance (§V-B) is measured on BM25, BGE, and a hybrid — three retrievers with overlapping but non-identical representational biases. A mutation can reliably exploit one of those biases (e.g., a lexical overlap that fools BM25, or an embedding-space artifact specific to bge_base) without exploiting a bias shared by E5 or GTE. The ≈24–25% held-out challenge rate for hard negative and conflict provides evidence for a cross-retriever component of these failures, alongside a larger construction-retriever-specific component that construction-time metrics alone cannot distinguish from it; §V-C's non-transfer taxonomy localizes most of that specificity to cases where the mutation reaches the held-out candidate set without dislodging the gold passage's rank, rather than to cases where the mutation is invisible to the held-out retriever. We do not claim this establishes a fully retriever-agnostic effect — our held-out set is two dense bi-encoders (E5, GTE), not a representative sample of retriever architectures — only that the effect reproduces on held-out retrievers that played no role in construction or filtering. This is precisely the decomposition RQ3 is designed to expose, and it is why we treat §V-B and §V-C as answering different questions rather than the same question measured twice.

The boundary case study is a compact illustration of the pipeline working end to end. A rule-based operator with a 100% automatic semantic pass rate looked, during development, like the safest of the five operators; an early human audit showed the opposite — systematic answer leakage across the two split fragments, which automatic review alone had no way to catch, since a leaked answer is by construction semantically coherent and topically on-target. We fixed the split logic and re-audited; the corrected operator now passes cleanly (§V-A), and its behavioral signature is genuine, not an artifact:

Recall@5 drops by 0.29–0.34 on both held-out retrievers — the largest degradation of any operator on this metric — while ΔMRR stays near zero on E5 and is unexpectedly slightly negative-signed on GTE (mutated MRR marginally exceeds clean). A plausible, though not directly measured, mechanism is that dividing the gold evidence across two fragments makes each fragment individually a weaker match for the query than the original consolidated passage, so the odds that either fragment lands in the top-5 fall even though, when one does land, it is not obviously ranked worse; we leave confirming this mechanism to future work. The practical lesson is at the level of metrics, not just pipeline stages: a single scalar "challenge rate" would call boundary the paper's strongest result, and a semantic-only or MRR-only view would have missed its degradation entirely — only by keeping ΔMRR and ΔRecall@5 separate (Definition 2) does boundary's distinct signature become visible, which is also why we treat it as a diagnostic operator (§V) rather than a headline result alongside hard negative and conflict.

The Random Hard-Negative comparison (§V-E) extends F2's dissociation to a setting where it is easy to miss: a baseline that a human audit rates almost as topically valid as our LLM-generated negatives (96% vs. 100% ok) is nevertheless substantially less behaviorally effective across the reported retrieval measures. Because the two operators share the same acceptance pipeline and differ only in generation source, this rules out an alternative explanation for the noise result — that the dissociation is specific to noise's unusual construction (adding content rather than altering it). The same gap between topical plausibility and behavioral effect appears within a hard-negative operator that most benchmark pipelines would treat as a single, homogeneous category.

Implication. Benchmark construction pipelines that generate synthetic retrieval failures — whether via LLM rewriting or rule-based transformation — should treat semantic validation (including a human-audit component) and behavioral validation as the two, separately reported gates that determine construction-time acceptance (Fig. 1), and should report cross-retriever transferability as a separate, held-out measurement rather than assuming it from acceptance. A mutation's semantic well-formedness, its construction-time effect, and its cross-retriever generalization are three different claims; collapsing them into one acceptance decision, or silently treating an accepted mutation as validated for transfer, makes at least one of these claims by assumption rather than by measurement.

VII. Limitations

- Corpus and domain: seeds are Wikipedia-derived (RAG-FailBench); results may not generalize to enterprise or domain-specific corpora.
- Scale: 531 unique seeds, short of the 1,000-seed target originally planned for this run.
- Boundary operator scope: even in its corrected form, boundary's behavioral signal is a genuine Recall@5 degradation rather than an MRR effect (§V-C, §VI); CR values for boundary are not directly comparable to hard negative/conflict without this caveat, and single-passage MRR may understate or mischaracterize fragmentation-type failures more generally.
- Noise yield: noise is semantically valid and illustrates RQ2 cleanly, but only 5/531 instances currently pass behavioral acceptance, too few to serve as a practical benchmark operator without redesigning the operator to raise yield.
- Scope: reranker recovery, generator EM/F1/faithfulness, and serving cost/latency are out of scope.
- Held-out retriever set: both held-out retrievers are dense bi-encoders; transfer to sparse or late-interaction retrievers is untested.
- Semantic audit annotation: the human audit (§III-B, §V-A) uses a single annotator; we therefore cannot report inter-annotator agreement for the 249/250 ok rate.
- Evaluation depth: held-out transfer outcomes (§V-C) are read from rankings truncated at top-20; a mutation's true held-out rank when it falls outside the top-20 is unknown, so the "never Top-20" bucket in Table V is a lower bound on invisibility, not an exact count.
- Threshold sensitivity: the analysis in §V-B covers a bounded range of ΔMRR cutoffs and two agreement rules, not an exhaustive search over possible behavioral-acceptance policies.
- Language: English only.

VIII. Conclusion

Synthetic retrieval failures have multiple validity dimensions: a mutation can be semantically valid yet behaviorally ineffective, or behaviorally effective on the retrievers used to construct it yet fail to transfer to held-out retrievers. This paper separates and evaluates all three properties empirically. On RQ1, automatic validation and independent human audit jointly establish semantic validity, with the corrected operators achieving 99.6% overall human-audit acceptance; a development-time boundary defect further shows why independent audit remains valuable. On RQ2, two independent operators — noise, and a Random Hard-Negative baseline compared against LLM-generated hard negatives — each show a large gap between semantic validity and behavioral effectiveness, ruling out an explanation specific to either operator's construction; an offline sensitivity check further shows this gap is not an artifact of the specific ΔMRR threshold used to define it. On RQ3, hard negative and conflict mutations that are behaviorally effective at construction time still only transfer to held-out retrievers in ≈24–25% of cases, a non-transfer taxonomy traces most of that gap to cases where the held-out retriever leaves the gold passage's rank undisturbed rather than to failed candidate retrieval, and boundary's superficially higher transfer rate turns out to be a ΔRecall@5 effect rather than the ΔMRR effect seen in the other operators — a further illustration that which metric is reported changes what "effective" means. Together these results argue that semantic validity, behavioral effectiveness, and cross-retriever

transferability should be measured as three separate properties, not collapsed into a single benchmark-quality judgment.

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